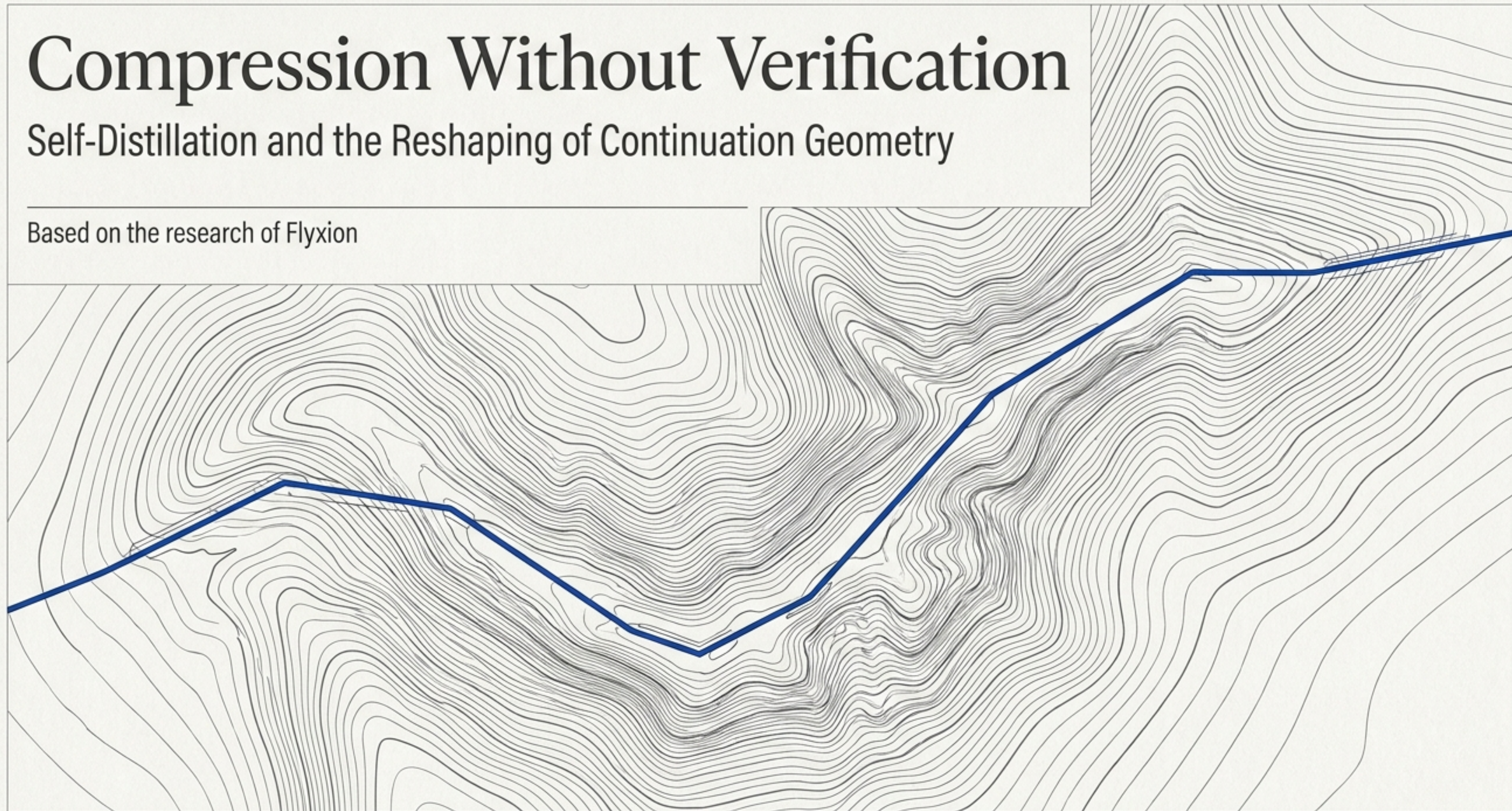


Compression Without Verification

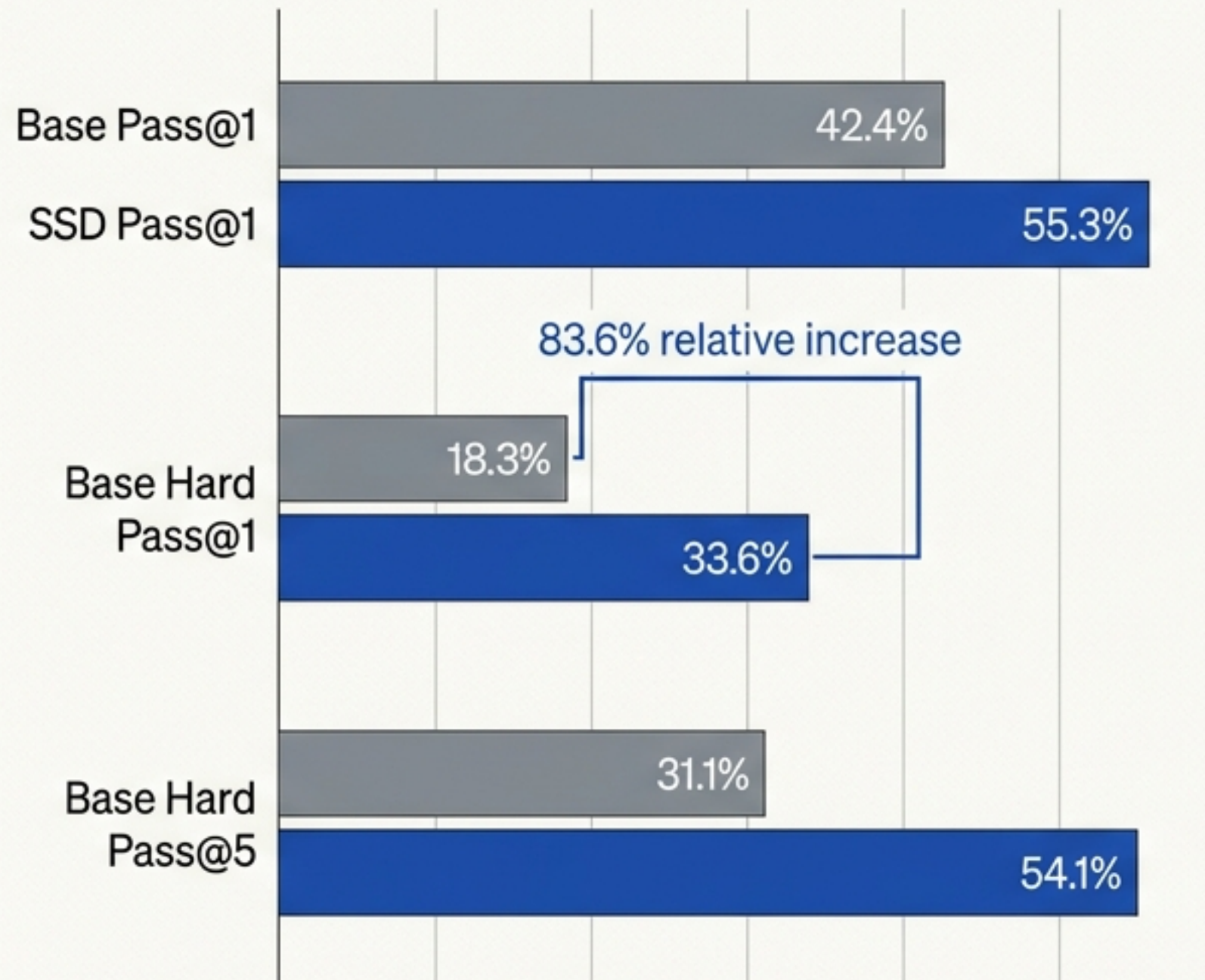
Self-Distillation and the Reshaping of Continuation Geometry

Based on the research of Flyxion



Models can improve at complex tasks without newly verified task information.

LiveCodeBench Performance (Qwen3-30B-Instruct)



- The base model was fed 10,000 competitive programming problems. For each, one answer was sampled.

□ ————— □

No answers were executed.

□ ————— □

No answers were tested.

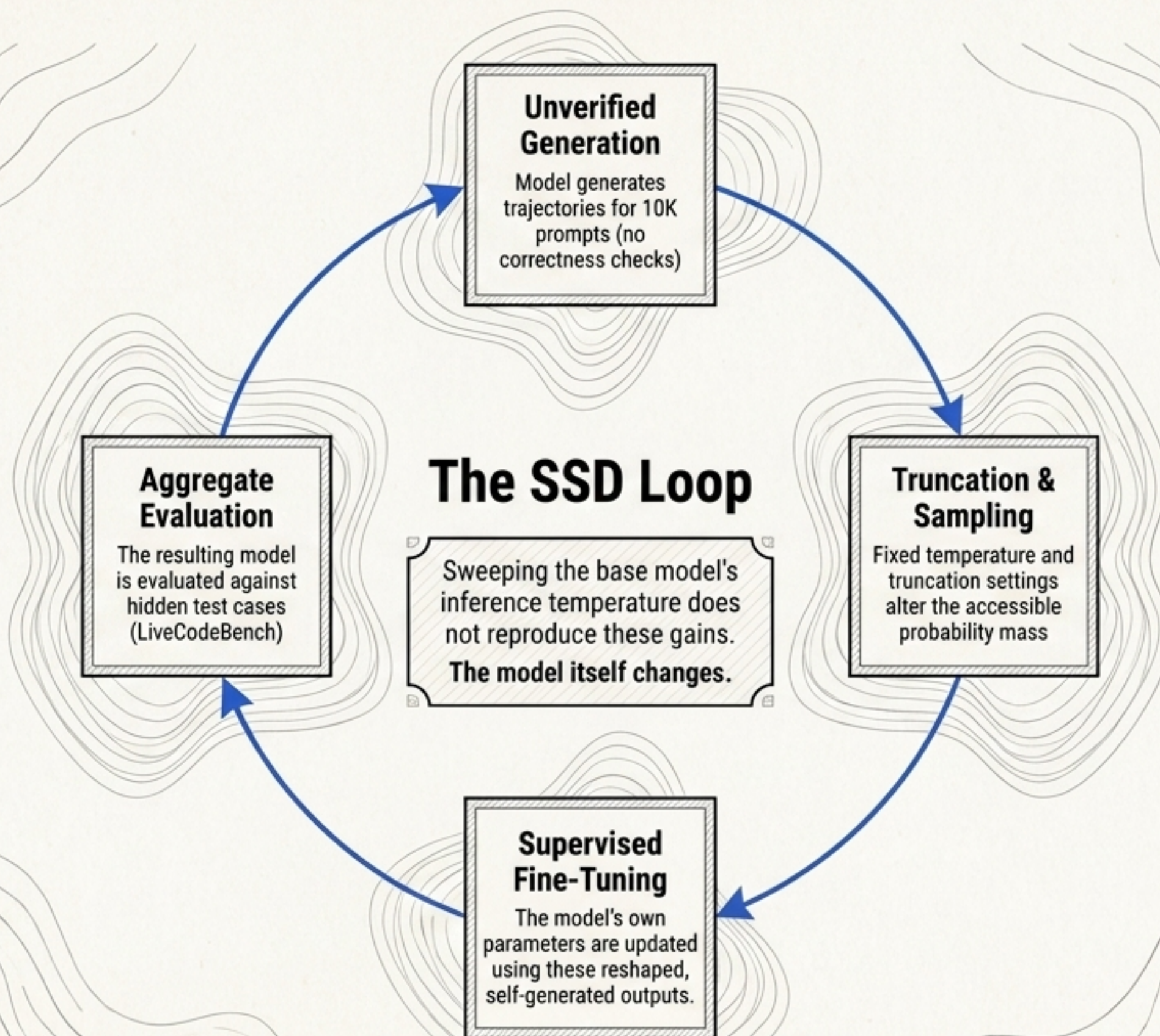
□ ————— □

No answers were filtered.

□ ————— □

The model was fine-tuned on this unverified self-generated corpus, yet achieved massive capability gains.

The Simple Self-Distillation loop creates improvement from internal distributions.



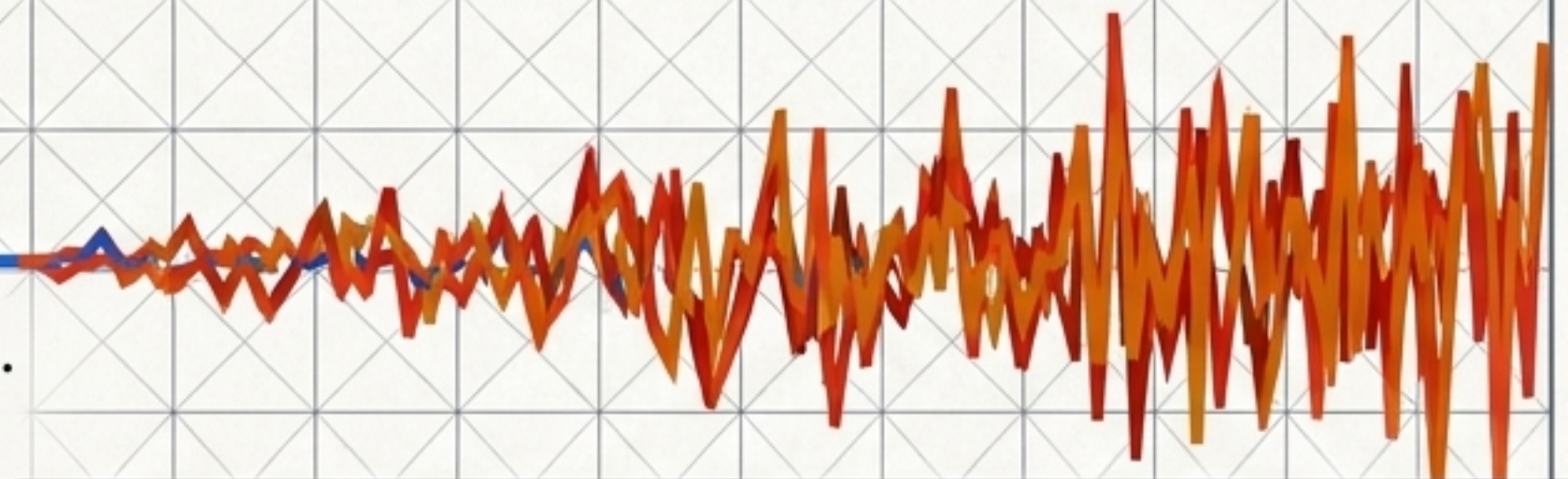
Complete program correctness is not required for the training signal to work.

Standard SSD

Standard training uses truncation.
Most retained code is syntactically coherent.
Results in maximum capability gain.

The Stress Test

Truncation is removed. Sampling temperature is raised.
The majority of training samples drift into incoherent, non-extractable multilingual output.



When fine-tuned on this heavily corrupted, largely non-executable data, the model still shows a substantial capability gain over the base model. How?

Tokens exist in specific topological environments: narrow **Locks** or expansive **Forks**.



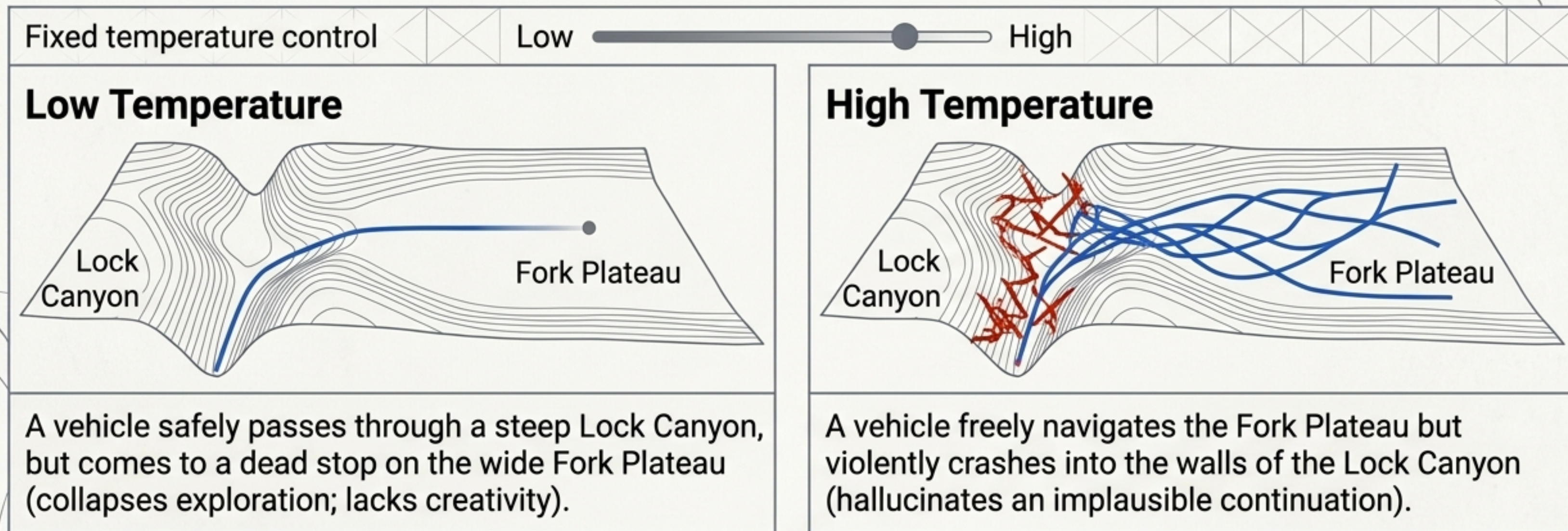
Locks



Forks

	Locks	Forks
Definition	A narrow continuation seam. Preceding context forces one overwhelmingly likely continuation (e.g., closing a nested structure).	A region of broad continuation width. Several distinct, genuinely viable continuations compete (e.g., choice of algorithmic strategy).
The Error Tail	The base model assigns non-zero probability to a long, implausible error tail .	Retaining multiple pathways is a feature of capability , not an error.
Post-SSD Behavior	Probability mass narrows further . The error tail is suppressed .	Broad distribution is preserved . Viable alternatives remain reachable.

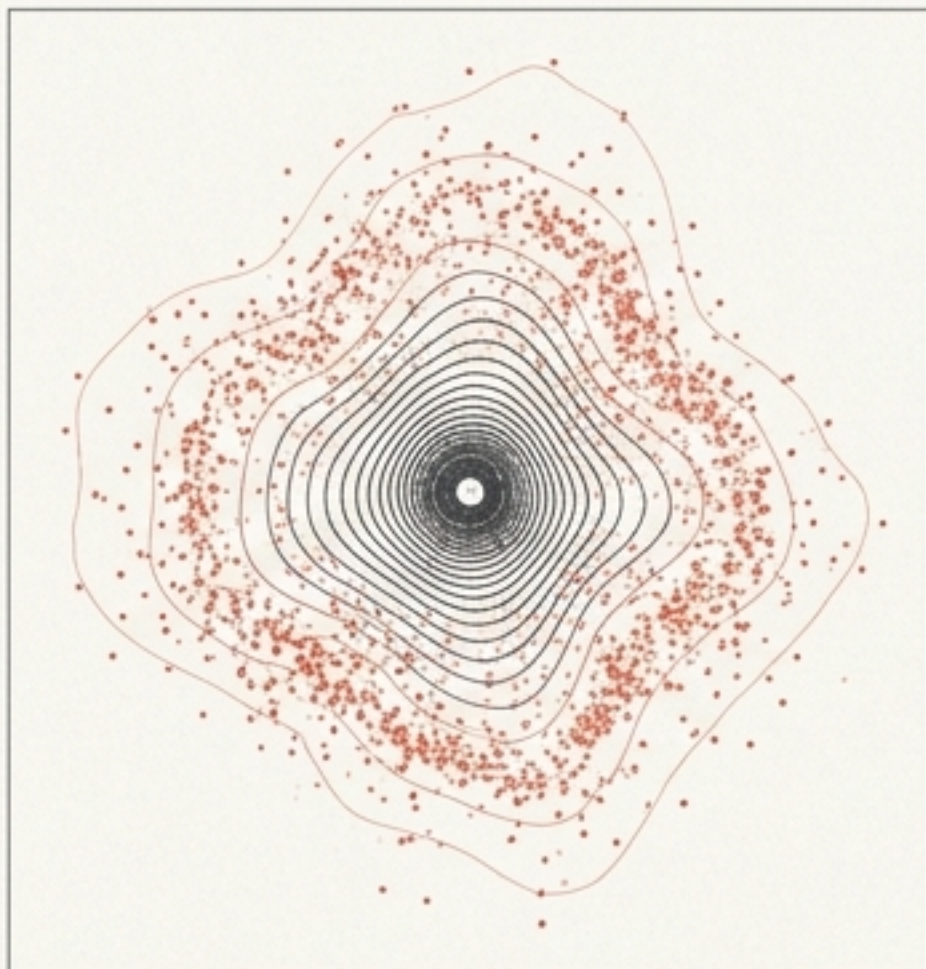
Single-parameter decoding policies fail at the Precision-Exploration conflict.



Core Insight: A one-parameter control policy is fundamentally mismatched with a continuation geometry that varies position-by-position.

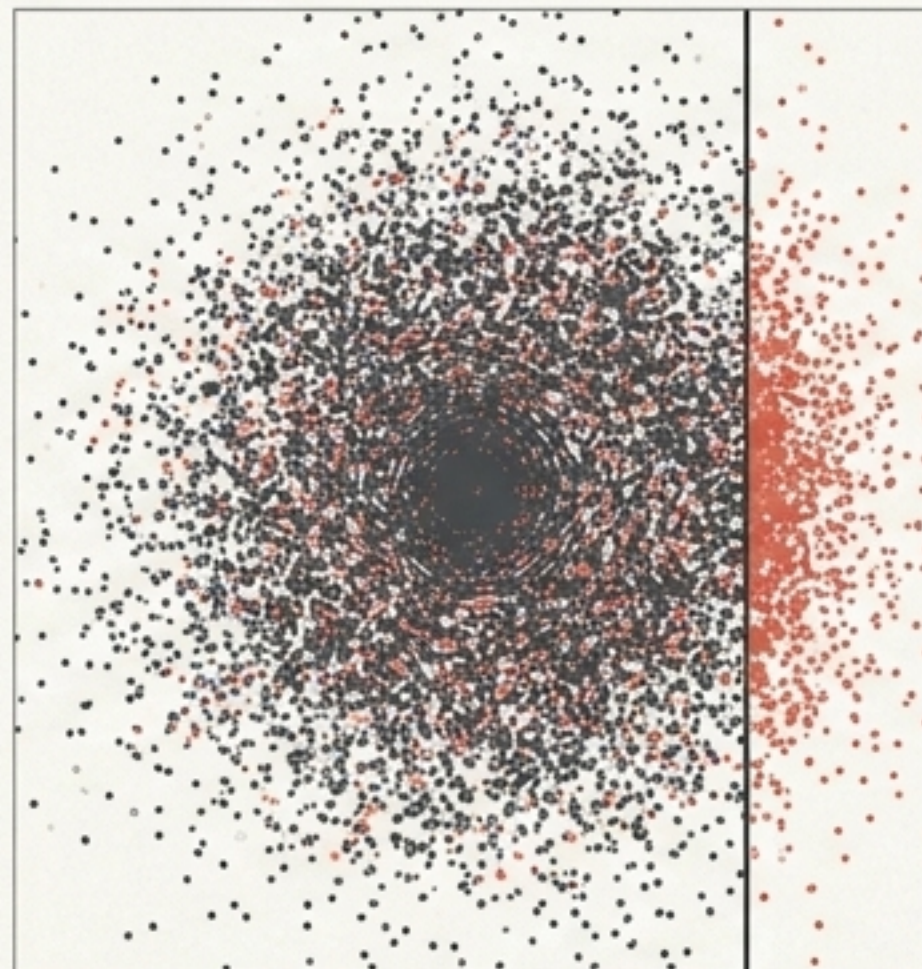
Self-distillation forces support compression and within-support reshaping

Part 1
The Base State



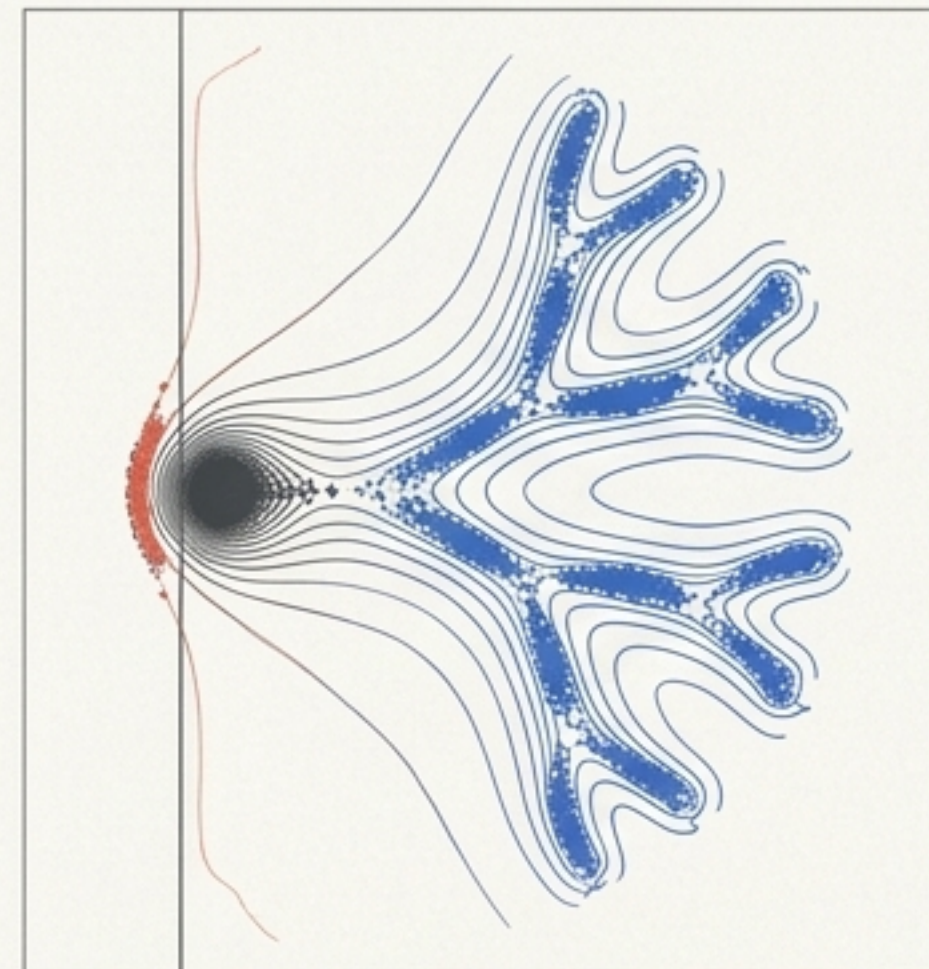
The original model contains viable strategies but wastes probability mass on diffuse error tails at lock positions.

Part 2
Expansion



Sampling at altered temperatures exposes alternative pathways. Truncation chops off the worst extremes.

Part 3
Non-Neutral Compression



Fine-tuning folds this expanded material back in. It relegates the low-ranked tail at locks and stabilizes the retained alternatives at forks.

The model does not learn new facts; it reshapes its continuation geometry.

The Illusion: Knowledge Creation

Hypothesis: The model taught itself new problem-solving capabilities without external signals.

Evidence Against: Gains are restricted to the training domain. Smaller models lose out-of-domain capability (a budget redistribution, not newly created competence).

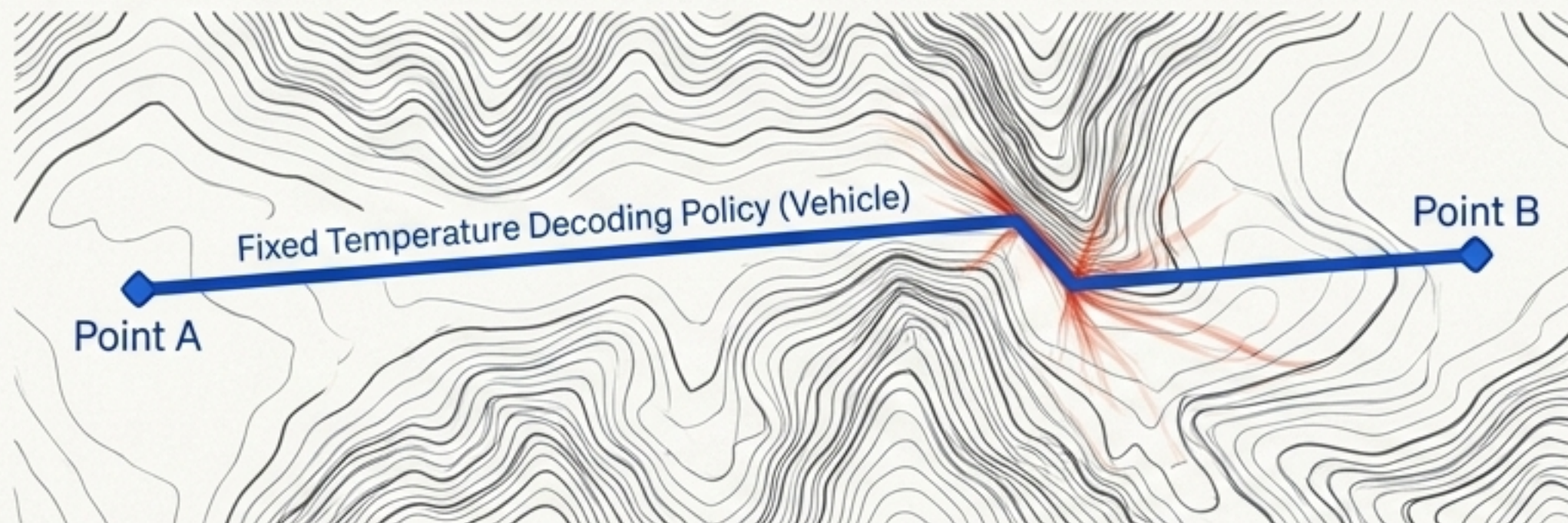
The Reality: Geometry Deformation

Hypothesis: The model already represented viable continuations at forks, and correct probabilities at locks. No single decoding policy could exploit both simultaneously.

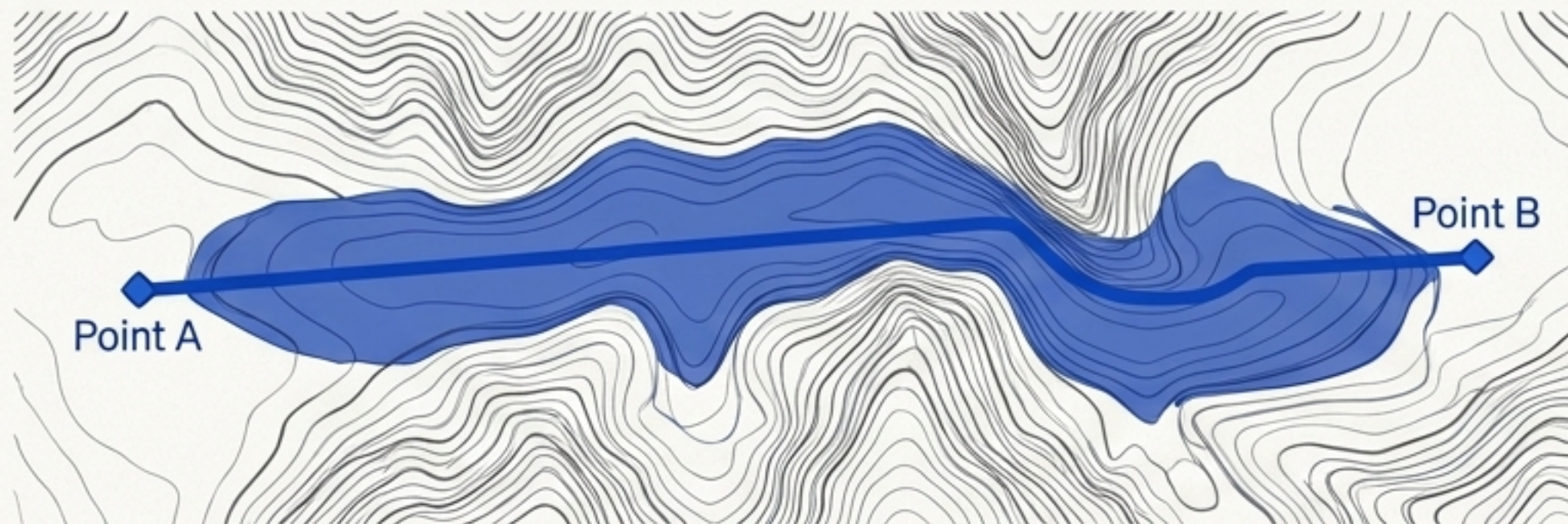
The Shift: SSD reshaped the internal landscape so a uniform policy could finally extract the capability already present in the weights.

The Admissible Region expands to allow a uniform decoding policy to succeed.

Before



After



Admissibility Across Substrates

The reachable set of correct programs grows not because raw representational knowledge expanded, but because the continuation geometry became compatible with a simple, fixed control budget. SSD bulldozes a path through the existing latent space.

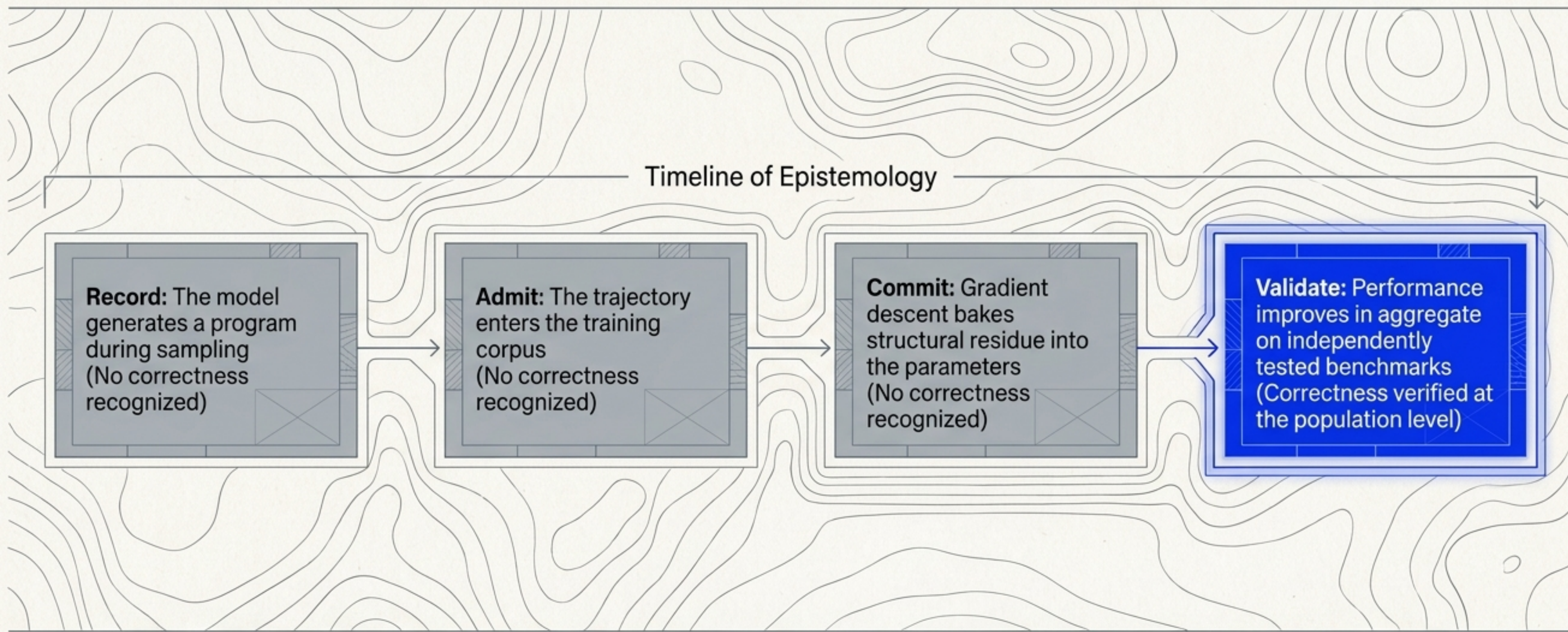
Corrupted data improves models because the structural residue is preserved.

Anatomy of a Failure



Arbitrary noise cannot be compressed into competence. But corrupted samples emitted by a trained model conditioned on meaningful problems retain dense local regularities. SSD compresses this structured residue. **The model learns the local syntax of the lock, even if the eventual macro-program fails.**

Verification is deferred to the aggregate population, not eliminated.



Takeaway: Optimization is separated from warranted belief. Improved aggregate accuracy does not certify any single output is correct.

The system is verification-free, but it is not information-free.

Verification-Free

- True: No correctness labels are attached to complete trajectories.
- True: Individual training trajectories are never validated before admission.

NOT Information-Free

Four sources of structure pre-exist the SSD update:

1. Pretrained regularities in the base model.
2. The highly specific competitive-programming prompts.
3. The base conditional distribution ranking the continuations.
4. The researcher-selected temperature and truncation policy.

Takeaway: What is missing is an external correctness judgment, not external structure.

Four precise experiments are required to isolate continuation geometry from generic adaptation.

Independent Topology Labeling

Classify Locks and Forks by semantic distinctness before measuring how SSD treats them (avoiding circular sharpness measures).

Corrupted-Data Decomposition

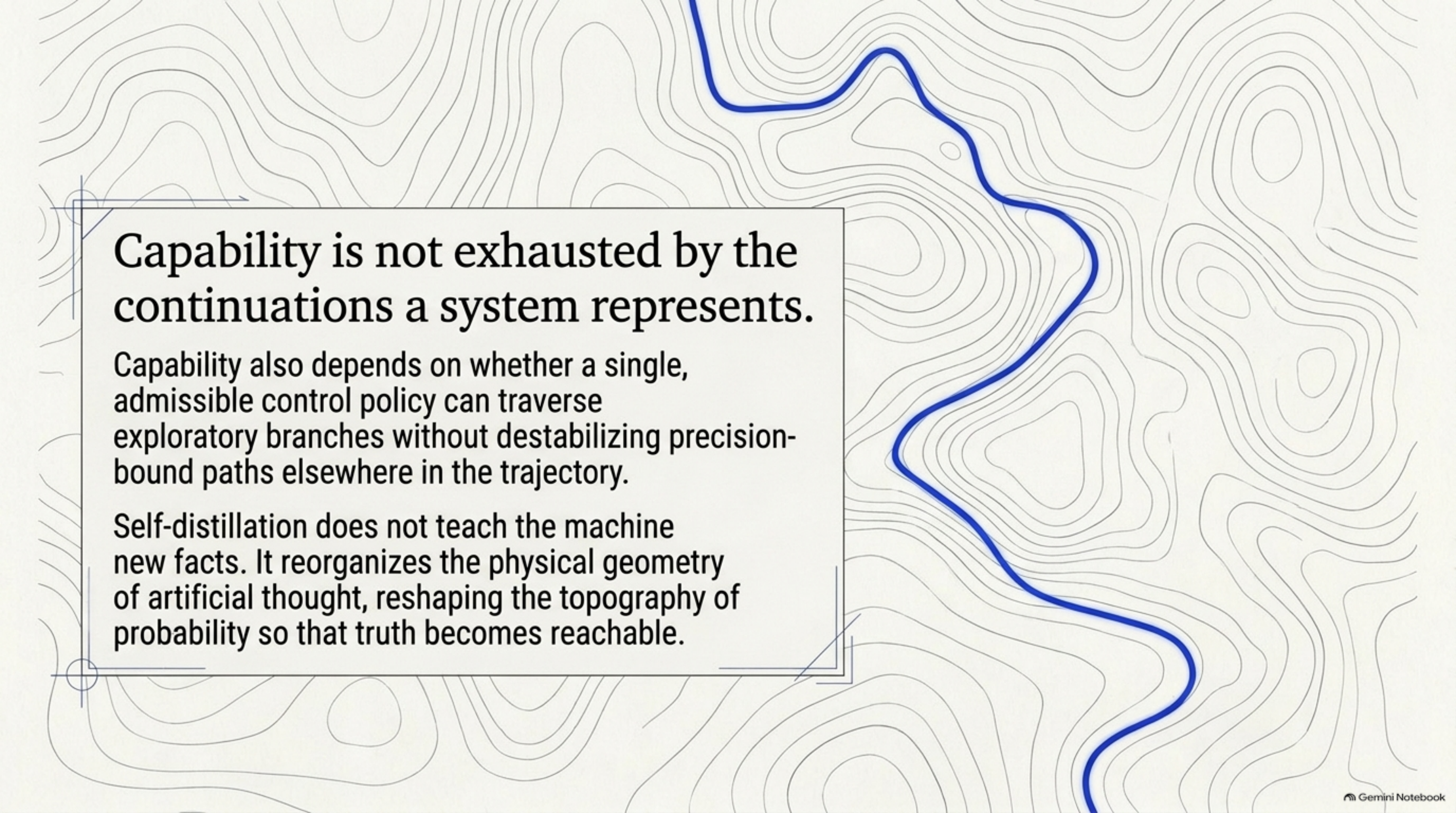
Train separately on coherent-prefixes-only, corrupted-suffixes-only, and fully shuffled tokens to isolate the source of the signal.

Direct Diversity Measurement

Cluster sampled programs by algorithmic strategy to prove Pass@5 gains are genuinely preserved branching structures, not just higher single-strategy reliability.

The Cross-Scale Bridge Test

A 4-way training comparison matching token counts across: correct solutions, syntactically valid (but wrong) solutions, truncated coherent prefixes, and prompt-matched unrelated code.

The background of the slide is a topographic map with light gray contour lines. A thick blue line, representing a path or trajectory, starts at the top center, moves right, then curves down and left, then down and right, then down and left, and finally down and right towards the bottom right corner. A white rectangular box with a thin gray border is positioned on the left side of the slide, containing three paragraphs of text. The box has a small blue circle in the bottom-left corner and a small blue line segment in the top-left corner.

Capability is not exhausted by the continuations a system represents.

Capability also depends on whether a single, admissible control policy can traverse exploratory branches without destabilizing precision-bound paths elsewhere in the trajectory.

Self-distillation does not teach the machine new facts. It reorganizes the physical geometry of artificial thought, reshaping the topography of probability so that truth becomes reachable.